

Algebra 8 Placement Test

Sienna Ash · 7th Grade — End of Year

Name: _____

Block: _____

Date: _____

~75 minutes · 100 points total. Show all work for full credit. Calculator allowed on Sections 3–5. No calculator on Sections 1–2.

Section 1 — Evaluating Expressions & Function Notation

14 pts

Q1.

2 pts

Evaluate $3x - 5$ when $x = 4$.

Answer: _____

Q2.

4 pts

Evaluate $2a^2 - 3b$ when $a = -2$ and $b = 5$.

Answer: _____

Q3.

4 pts

Evaluate $\frac{5x - y}{2}$ when $x = 3$ and $y = -1$.

Answer: _____

Q4.

4 pts

Let $f(x) = 3x - 1$.

(a) Find $f(-2)$.

2 pts

$$f(-2) = \underline{\hspace{2cm}}$$

(b) Find the value of x where $f(x) = 11$.

2 pts

$$x = \underline{\hspace{2cm}}$$

Section 2 — Solving Equations

30 pts

No calculator. Show every step.

Q5.

4 pts

Solve: $2x + 9 = 21$

$$x = \underline{\hspace{2cm}}$$

Q6.

4 pts

Solve: $3(x - 4) = 15$

$$x = \underline{\hspace{2cm}}$$

Q7.

6 pts

Solve: $4x - 7 = x + 8$

$$x = \underline{\hspace{2cm}}$$

Q8.

6 pts

Solve: $2(3x + 1) = 4x - 6$

$x =$ _____

Q9. 4 pts

Solve: $\frac{2}{3}x - 4 = 6$

$x =$ _____

Q10. 3 pts

How many solutions does $5x + 2 = 5x - 3$ have? Show your work and explain.

Number of solutions: _____

Q11. 3 pts

How many solutions does $4(x - 1) = 4x - 4$ have? Show your work and explain.

Number of solutions: _____

Section 3 — Proportional Relationships

18 pts

Calculator allowed.

Q12. 6 pts

Given the table:

x	0	2	4	6
y	3	7	11	15

(a) Is this a proportional relationship? Explain.

2 pts

(b) What is the y-intercept?

2 pts

y-intercept: _____

(c) Write the equation in $y = mx + b$ form.

2 pts

Equation: _____

Q13.

6 pts

A line passes through $(0, -2)$ and $(4, 6)$.

(a) Is this proportional? Why?

2 pts

(b) What is the slope?

2 pts

$m =$ _____

(c) Write the equation.

2 pts

Equation: _____

Q14.

6 pts

A parking garage charges a flat \ entry fee plus \ per hour.

(a) Write an equation for total cost C based on hours h .

2 pts

$C =$ _____

(b) What does the slope represent in this context?

2 pts

(c) Is this a proportional relationship? Why?

2 pts

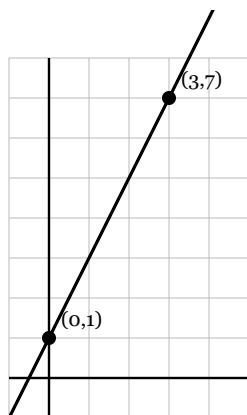
Section 4 — Identifying Slope

20 pts

Q15.

4 pts

The graph shows a line through $(0, 1)$ and $(3, 7)$. Find the slope. Show your work.

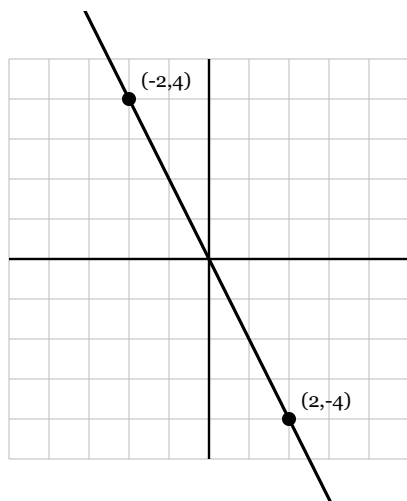


$m =$ _____

Q16.

4 pts

The graph shows a line through $(-2, 4)$ and $(2, -4)$. Find the slope. Show your work.



$m =$ _____

Q17.

6 pts

Find the slope of the line through $(-1, 5)$ and $(3, -3)$. Show all work.

$m =$ _____

Q18.

3 pts

What is the slope of $y = -4x + 7$?

$m =$ _____

Q19.

3 pts

What is the y-intercept of $y = 3x - 5$?

$b =$ _____

Section 5 — Graphing Lines

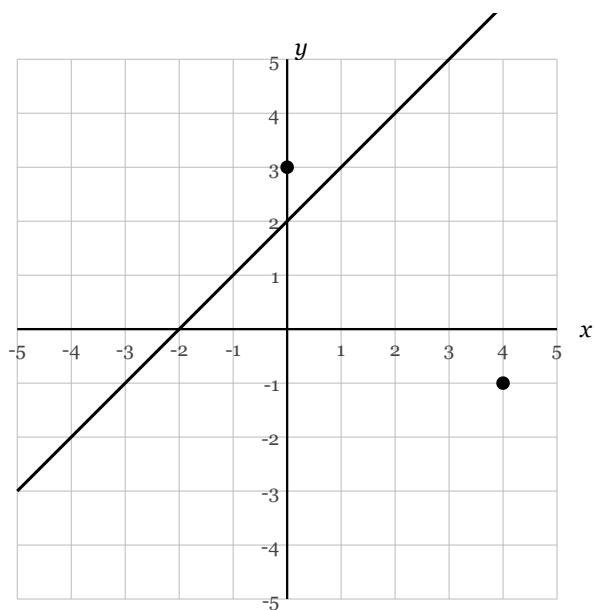
18 pts

Pre-printed coordinate grids below. Use them.

Q20.

6 pts

The line below passes through $(0, 3)$ and $(4, -1)$.



(a) What is the slope?

2 pts

$m =$ _____

(b) What is the y-intercept?

1 pt

$b =$ _____

(c) Write the equation.

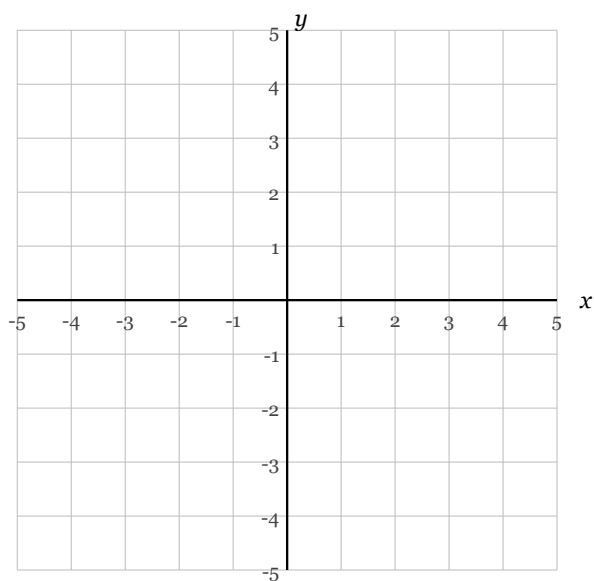
3 pts

Equation: _____

Q21.

6 pts

Graph $y = 2x - 3$ on the coordinate grid below.



Label each point you plot with its coordinates.

(a) Plot and label the y-intercept.

2 pts

(b) Use the slope to plot and label two additional points.

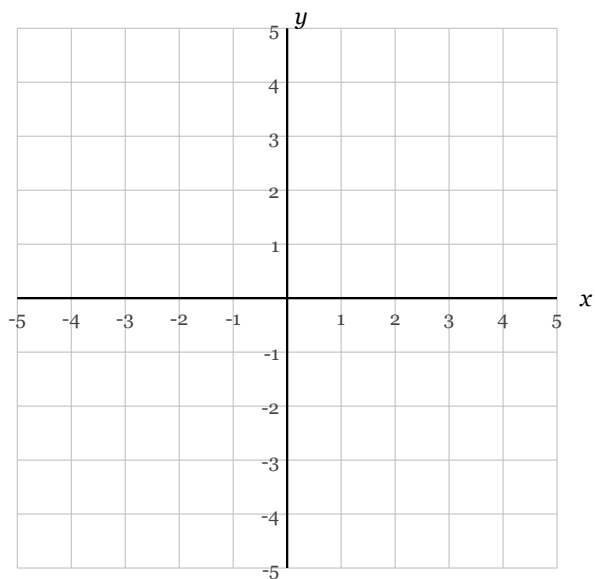
4 pts

Q22.

6 pts

The table shows a linear relationship:

x	-2	0	2	4
y	1	5	9	13



(a) Plot all four points and draw the line.

3 pts

(b) Write the equation of the line.

3 pts

Equation: _____